

Jyothimanoahar Pothireddy

AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of experience designing, developing, and deploying end-to-end production-grade AI/ML and Generative AI solutions that deliver measurable business impact across insurance, telecom, banking, and pharma. Owns the full ML lifecycle — data engineering, feature engineering, model training, evaluation, deployment, and monitoring — with deep expertise in Python, PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, and SQL. Builds LLM-powered applications, RAG architectures, agentic AI workflows, and multi-agent systems using LangChain, LangGraph, LlamaIndex, and Hugging Face. Deploys scalable AI systems on AWS (SageMaker, Lambda, ECS/EKS), Azure (AI Foundry, Azure ML), and GCP (Vertex AI) using Docker, Kubernetes, and CI/CD pipelines. Implements MLOps best practices — MLflow, DVC, experiment tracking, model versioning, drift monitoring, and A/B testing. Applies Responsible AI principles including SHAP/LIME explainability, bias detection, model governance, and compliance. Strong cross-functional collaborator who translates business needs into scalable technical solutions and communicates complex AI results clearly to technical and non-technical stakeholders.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Bash/Shell Scripting

Machine Learning: Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Anomaly Detection, Time Series Forecasting, Recommendation Systems, NLP, Feature Engineering, A/B Testing, Causal Inference, Statistical Modeling, Experiment Design, Hyperparameter Tuning, Model Evaluation, Cross-Validation, Bias/Variance Analysis, Error Analysis, Responsible AI, SHAP, LIME, Explainable AI (XAI), Bias Detection, Model Governance

Deep Learning & Neural Networks: Python, PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, CatBoost, Pandas, NumPy, SciPy, ONNX, CUDA, GPU Acceleration, CNNs, RNNs, LSTMs, Transformers, Distributed Training

Generative AI & LLMs: LLMs (GPT-4, Claude, Gemini, Llama), Generative AI, RAG Architectures, Agentic AI, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, Semantic Kernel, MCP Server, Prompt Engineering, Fine-Tuning (PEFT, LoRA), Hugging Face, OpenAI API, Vector Databases (FAISS, Pinecone), Semantic Search, Embedding Pipelines, Knowledge Graphs, LLM Evaluation (RAGAS, DeepEval, LangSmith), Hallucination Detection, Guardrails, Responsible AI

Natural Language Processing (NLP): NLTK, spaCy, NER, Semantic Search, Text Classification, Text Summarization

Vector Databases & Search: FAISS, Pinecone, ChromaDB, Weaviate, Embeddings, Vector Search, Semantic Search

MLOps & Model Deployment: MLOps, LLMops, MLflow, DVC, Kubeflow, Docker, Kubernetes (ECS/EKS), CI/CD Pipelines, GitHub Actions, Terraform, AWS CDK, Model Versioning, Experiment Tracking, Drift Monitoring, A/B Testing, Model Registry, Feature Store, FastAPI, Flask, REST APIs, gRPC, Real-Time & Batch Inference, Model Monitoring, Observability, Weights & Biases

Cloud Platforms: AWS (SageMaker, Bedrock, Lambda, ECS/EKS, EC2, S3, API Gateway, Step Functions), Azure (AI Foundry, Azure ML, Azure OpenAI), GCP (Vertex AI, BigQuery), Databricks, Snowflake, DBT, Fivetran

Data Engineering & Big Data: Apache Spark, Apache Kafka, Apache Airflow, SQL, PostgreSQL, MongoDB, ETL/ELT Pipelines, DBT, Databricks (Medallion Architecture), Snowflake, BigQuery, Data Lakes, Data Warehousing, Feature Engineering, Data Quality, Data Versioning

Databases: MySQL, PostgreSQL, MongoDB, Cassandra, NoSQL Databases

Data Analysis & Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Pandas, NumPy, SciPy, Statistical Analysis, Dashboard Development, Data Storytelling

Responsible AI: Explainable AI (XAI), SHAP, LIME, Fairlearn, AI Governance, Bias Detection, Model Interpretability, Responsible AI, Model Risk Management, Compliance (SOC 2, ISO 27001), Audit-Ready Documentation

Methodologies & Tools: Python, SQL, Git, GitHub, GitLab, Agile, Scrum, Claude Code, Cursor, GitHub Copilot, Cross-Functional Collaboration, Technical Mentorship, Code Reviews, Documentation, End-to-End Ownership

PROFESSIONAL EXPERIENCE

AI / Machine Learning Engineer

American Income Life Insurance Company | Waco, TX

July 2025 – Present

- Built an end-to-end risk intelligence platform covering claim risk scoring, fraud detection, and policyholder retention — training XGBoost, LightGBM, and PyTorch models on 1M+ policy records, running cross-validation and error analysis to validate production readiness, and improving model precision/recall by 18% against baseline.
- Designed and shipped a document intelligence system for policy and claims processing — fine-tuned domain-specific LLMs using PEFT/LoRA, built a RAG pipeline with FAISS vector search and LangChain orchestration, and deployed an agentic workflow that automated triage across 50K+ documents, reducing manual review effort by 45%.
- Architected a multi-agent AI system with specialized agents for data retrieval, planning, and task execution — implementing agent memory management, tool-use patterns via MCP Server, and LangSmith tracing; ran RAGAS and DeepEval benchmarks to catch hallucinations and enforce guardrails before production rollout, improving knowledge retrieval speed by 60%.
- Deployed the full model serving stack on AWS SageMaker and ECS/EKS — containerizing inference services with Docker, exposing them through FastAPI REST and gRPC endpoints, and configuring CloudWatch-based observability and alerting; achieved 99%+ uptime with sub-second p99 latency in production.
- Built the team's feature engineering platform on Databricks (bronze/silver/gold medallion architecture) — orchestrating Spark and Airflow pipelines that transformed 2M+ raw operational records into model-ready features, tracked with MLflow and DVC, and monitored for data drift; cut feature preparation time by 35% and made experiments fully reproducible.
- Established the team's MLOps/LLMOps framework from scratch — authoring Terraform infrastructure-as-code, configuring GitHub Actions CI/CD for automated model evaluation and deployment, maintaining a central model registry, and embedding SHAP/LIME explainability and SOC 2-aligned governance documentation into every release; reduced deployment cycle time from days to hours.
- Served as the AI/ML technical liaison between engineering, product, and business teams — mentoring junior data scientists, leading weekly model review sessions, and translating model outputs into executive-facing Power BI dashboards that surfaced 20+ AI-driven KPIs on claims, retention, and risk trends.

Machine Learning Engineer

Dec 2024 – Jun 2025

Crown Castle | Houston, TX

- Developed a suite of 6+ production ML systems for network asset health — including an outage prediction model using Transformer-based time series, an anomaly detection system for equipment monitoring, and a capacity forecasting pipeline using XGBoost; each system was trained, validated, and deployed directly into operations workflows.
- Built the real-time feature engineering pipeline powering all network ML models — ingesting 500K+ daily telemetry events via Kafka into Spark, materializing features to S3, and serving models through gRPC endpoints on Kubernetes; implemented CloudWatch dashboards and PagerDuty alerting for production incident response.
- Led a structured A/B testing program across all model improvements — defining success metrics, running ablation studies, and using causal inference to isolate the true lift of each change; added SHAP-based explainability dashboards that gave field engineers visibility into model decisions, reducing manual review effort by 30%.
- Built an internal engineering knowledge assistant — indexing field documentation into a FAISS vector store, building a retrieval-augmented generation pipeline with LangChain and LlamaIndex, and deploying an agentic interface that let field engineers query complex troubleshooting guides in natural language; reduced documentation search time by 50%.
- Owned the MLOps platform for all telecom ML models — containerizing services with Docker, deploying on Kubernetes with Terraform-managed infrastructure, tracking experiments in MLflow and Weights & Biases, and enforcing PyTest unit/integration coverage gates in GitHub Actions CI/CD; reduced rollback incidents by 40% through disciplined versioning and testing.
- Partnered with network operations leadership to translate infrastructure reliability goals into concrete ML problem statements; presented model performance findings and tradeoff analyses to senior stakeholders; authored Responsible AI documentation and model cards for all production systems to meet enterprise compliance standards.

Data Scientist

Jun 2022 – Jun 2024

Indian Bank | Chennai, India

- Built the credit and fraud ML stack across four model families — credit risk scoring, loan default prediction, transaction fraud detection, and customer churn — trained on 3M+ banking records using XGBoost and LightGBM; ran rigorous A/B tests to validate each model's production lift and improved AUC by 20% over legacy rule-based systems.

- Designed the feature store for banking ML — pulling from AUM, transaction history, CRM, and bureau data using DBT transformations and SQL, materializing time-windowed behavioral features with strict data leakage controls; added SHAP explainability and bias assessment reports that the risk governance team used in model approval reviews.
- Automated the bank's regulatory reporting and model retraining pipelines — orchestrating Spark and Airflow ETL across 10+ core banking systems into Snowflake, triggering automated drift detection alerts and retraining jobs via CI/CD; reduced reporting cycle time by 40% and ensured models stayed calibrated as portfolio composition shifted.
- Led the data science practice for the analytics team — mentoring two junior analysts, conducting bi-weekly code reviews, and building executive Power BI dashboards that replaced 10+ hours of manual reporting per week by surfacing 25+ real-time KPIs on portfolio risk, customer behavior, and credit performance.
- Embedded Responsible AI into every model release — running SHAP and LIME analyses to identify and mitigate bias, producing audit-ready model cards and governance documentation, and contributing to the bank's internal AI policy framework to ensure all production models met RBI compliance expectations.

Data Scientist

Dec 2021 – May 2022

Apex Laboratories Pvt. Ltd.

| Chennai, India

- Built a demand intelligence platform for pharmaceutical supply chain — training XGBoost and PyTorch forecasting models on multi-year sales and inventory data, validating with time-series cross-validation, and deploying predictions to the planning system that drove procurement decisions; reduced stockout and overstock risk by 15%.
- Consolidated fragmented pharmaceutical data from 8 source systems — manufacturing, ERP, lab LIMS, and third-party suppliers — into a unified Spark and Airflow ETL layer feeding downstream ML models; implemented DBT transformations and data quality checks to ensure freshness and consistency across production pipelines.
- Delivered production forecasting and statistical models covering distribution scheduling and lead-time variability — applying LSTM and Prophet time series architectures on supply chain data, tracking experiments in MLflow, and reducing lead-time variability by 20% through model-driven scheduling recommendations.
- Developed an LLM-powered regulatory document assistant for the pharma team — fine-tuning a domain-specific model using PEFT/LoRA on clinical and regulatory corpora, building a RAG pipeline with LangChain and RAGAS evaluation, and deploying it internally to automate document classification and extraction; cut regulatory review time by 35%.
- Designed self-service analytics dashboards in Tableau and Power BI that gave commercial and operations teams real-time visibility into 20+ supply chain and sales KPIs — replacing weekly manual Excel reports and enabling same-day decision-making on inventory and distribution priorities.
- Standardized the pharma team's model development lifecycle — introducing MLflow experiment tracking, Docker containerization, and GitHub Actions CI/CD from scratch, establishing a model registry and monitoring framework, and using AI-assisted coding tools (Claude Code, Cursor) to accelerate development velocity; reduced production handoff time by 30%.

EDUCATION

Master of Science — Computer and Information Sciences

Texas, USA

Texas Tech University

CERTIFICATIONS

- **Google Cloud Certified — Professional Machine Learning Engineer** — Google Cloud
- **IBM AI Engineering Professional Certificate** — IBM / Coursera
- **Deep Learning Specialization** — DeepLearning.AI
- **AWS Certified Cloud Practitioner (Foundational)** — Amazon Web Services